F78PB Practical Test Solutions 2022

1.

```
> sample1=scan()
1: 70.1 83.0 124.7 93.0 73.0 218.3 99.1 128.6 87.4 80.7 67.7 59.5
13: 158.1 79.2 127.3 94.4 74.4 73.3 85.5 51.0 69.4 104.7 84.0 215.9
25: 78.6 98.1 56.7 76.3 97.8 103.0
31:
Read 30 items
> sample2=scan()
1: 72.5 85.9 129.0 96.2 75.5 225.8 102.5 133.0 90.4 83.5 70.0 61.5
13: 163.6 81.9 131.7 97.7 77.0 75.8 88.4 52.8 71.8 108.3 86.9 223.3
25: 81.3 101.5 58.7 78.9 101.2 106.6
31:
Read 30 items
> mean(sample1)
[1] 97.09333
> mean(sample2)
[1] 100.44
>
```

2 marks for input and calculation of the means of the samples. No need to print screen.

Since we know that the variance is 100, the formula for estimator 2 is better. It appears to be unbiased.

1 mark

This ties in with our theory because we have shown in class that estimator 2 is an unbiased estimator of sigma-squared from the sample variance.

1 mark

Alternative solution:

Show MSE 1 > MSE 2, hence estimator 1 is better.

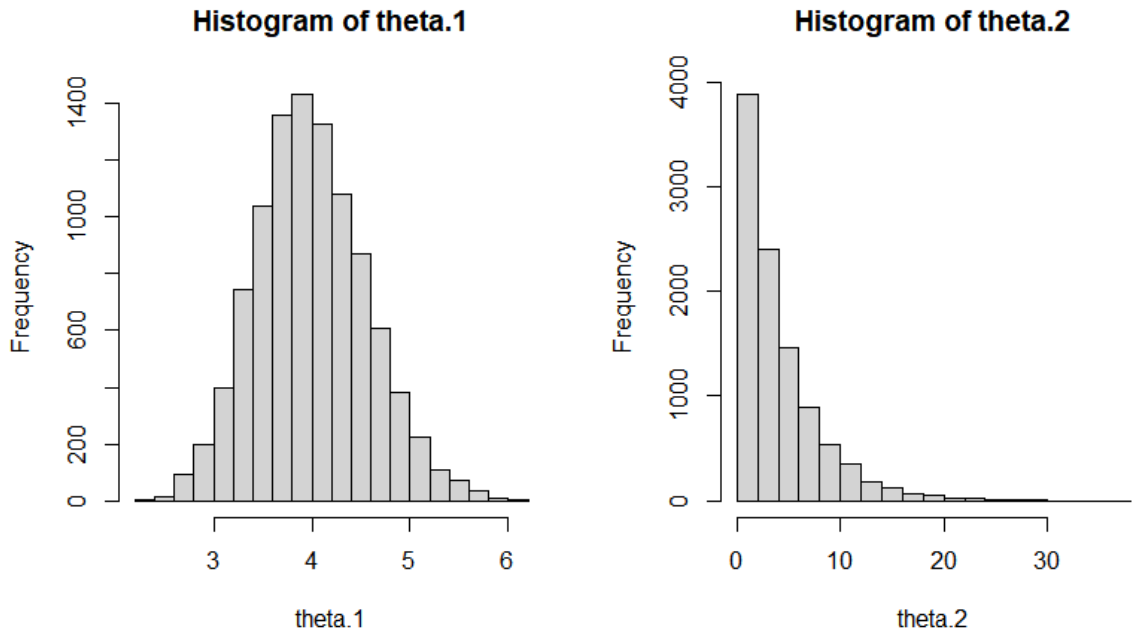
2. For my particular simulation, the summary statistics are as follows.

(a)

```
> summary(theta.1)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  2.322   3.602   3.961   3.992   4.359   6.029
> summary(theta.2)
    Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
 0.00008  1.16708  2.81319  4.02700  5.59614 36.33820
> var(theta.1)
[1] 0.3169057
> var(theta.2)
[1] 15.89453
```

1 mark for each summary, 1 mark for both variances (0.5 per variance)

(b)



1 mark graphs

The distribution of theta.1 seems to be that of a normal distribution 1 mark whilst theta.2 looks like an inverse curve such as 1/x or exp(-x) 1 mark.

(c) We have simulated values in R with theta = 4. To work out the bias for our sample we calculate the sample mean and subtract 4.

```
> mean(theta.1)-4
[1] -0.008428024
> mean(theta.2)-4
[1] 0.02700273
>
```

Both estimators seem quite accurate and are probably both unbiased, but the first estimator seems to give a better result.

1 mark

The MSE is calculated as bias^2 + var. The results are seen below.

```
> (mean(theta.1)-4)^2+var(theta.1)
[1] 0.3169768
> (mean(theta.2)-4)^2+var(theta.2)
[1] 15.89525
>
```

The MSE is much lower for the first estimator. 2 marks

(d) The sample size is 50…. Same as for our simulations. Use theta.1 to estimate the parameter as it has a lower MSE than theta.2 1 mark

This just involves taking the 1/(mean of our sample results),

where the mean of our sample results is 8.028. 1 mark

Hence, final answer is 0.125 (3 d.p.) 1 mark